Trails — Product Requirements Document

v0.2.0 · A Living Spatial Journal

Status: Implementation specification
Platform: Native iOS
Foundation: Extensa's existing indoor capture, 3D representation, 2D plan, and footsteps
Release focus: A polished personal experience spanning capture, memories, replay, place collections, and geographic discovery

> Capture a place through your eyes. Leave moments along the way. Return to the journey.

1. Product vision

Trails connects people, places, and personal memory.

A trail is not merely a room scan or GPS recording. It is an authored journey containing a spatial capture, a recorded path, and moments: photographs, spoken observations, written notes, or short videos.

The long-term vision is a community-built record of places through the people who experienced them.

Version 0.2.0 must first be valuable to one person, without publishing anything.

The user's immediate value

User need	Trails' response
"I want to remember being here."	Capture the place and what mattered along the way.
"I want to return to that experience."	Revisit the scene, path, photos, and voice recordings together.
"I have several perspectives on this place."	Keep multiple independent trails grouped under one place.
"Where have I left trails?"	Explore the collection geographically and by date.
"I want to show someone."	Open an uncluttered presentation experience on the device.

The core principle is **continuity**: capture, review, journal, map, and replay are different views of the same trail—not separate products.

2. Release boundaries

Included

- Private, on-device trails.
- Indoor spatial capture using the working Extensa implementation.
- Recorded movement and footsteps.
- Photo, audio, text, and short-video moments.
- Reliable saving and reopening.
- Journal and map views.
- Place collections containing multiple trails.
- Date, year, place, and text filtering.
- 3D and 2D exploration.
- Moment-by-moment replay.
- Clean presentation mode.
- Local creator identity.
- Starting another trail at the same place.
- Optional “Inspired by” relationships between local trails.

Not included

- Public directories or community publishing.
- Shareable internet links.
- Logos uploads or retrieval.
- Google/account synchronisation.
- AI-generated films, collages, or narration.
- AI navigation.
- Turn-by-turn guidance.
- Automatic indoor localisation in another person’s scan.
- Automatic merging of independent captures.
- Background spatial recording.
- Followers, ratings, points, or territory ownership.

These remain roadmap items. The interface must not display them as functioning features.

3. Product objects

```
Local profile
  └── Places
       └── Trails
            ├── Spatial capture
            ├── Recorded path
            └── Moments
                 ├── Photo
                 ├── Audio
                 ├── Video
                 └── Text
```

Place

A location grouping multiple trails.

A place may have:

- Name.
- Locality.
- Optional geographic anchor.
- Optional cover.
- Multiple independent captures and trails.

A place does not imply ownership or administrative authority.

Trail

One authored recording session.

Contains:

- Creator.
- Title and optional introduction.
- Place association.
- Optional floor.
- Capture assets.
- Recorded path.
- Ordered moments.
- Creation and modification dates.
- Recording and processing state.

Moment

A meaningful point in the journey.

It may contain multiple media types. A photograph and its spoken explanation can belong to one moment rather than appearing as unrelated records.

4. UX principles

Start quickly

No title, description, location form, or profile completion should block recording.

Preserve first, organise second

Save the capture before asking the user to polish its presentation.

Make the result recognisable

Use personal photographs, introductions, and voice recordings to give the trail character. Geometry supplies context, not the entire emotional experience.

Never lose context

Switching Journal/Map, 3D/Plan, or Moments/Explore must preserve the selected trail and relevant filters.

Make every action resolve clearly

Users should know:

- Whether recording is active.
- Whether media was saved.
- Whether processing is still underway.
- Whether a trail exists safely on the device.

Reveal complexity progressively

Simple actions stay visible. Editing, metadata, and advanced options belong in contextual sheets.

5. Application structure

Main destinations

Destination	Purpose
My Trails	Journal/map collection, search, places, and recording entry point
You	Local identity and personal contribution overview
Settings	Privacy, permissions, storage, diagnostics, and app information

Settings may live within You rather than occupy a tab.

Use a prominent “Start a trail” action. Do not require a dedicated recording tab.

Primary flow

```
My Trails
   ↓
Start a trail
   ↓
Capture + leave moments
   ↓
Finish
   ↓
Save and process
   ↓
Review
   ↓
Revisit or return to collection
```

6. First-run experience

Welcome

Headline:

> “Every place holds more than one story.”

Supporting copy:

> “Keep the places you experience, and the moments you want to remember.”

Actions:

- Start a trail.
- Explore a clearly labelled sample, if one exists.

Local identity

Ask for a display name at a non-blocking point.

Optional:

- Portrait.
- Short introduction.
- Interests.

Requirements:

- No email or password.
- No claim of verified identity.
- No mandatory interests.
- No permission wall.

Explain:

> “Your trails stay on this iPhone. Nothing is published automatically.”

The app must separately document its actual operating-system backup behaviour.

7. Capture experience

7.1 Entry points

The user can start:

- From My Trails.
- From a place collection.
- From an existing trail using “Start another trail here.”

Starting from a place preselects its association. It does not reuse or align a scan automatically.

7.2 Preparation

Show only what is necessary:

- Hardware/support check.
- Contextual camera permission.
- Brief capture instruction.

Suggested copy:

> “Move through the space. Leave a moment wherever something matters.”

Do not require GPS acquisition before indoor capture.

7.3 Active recording

Main canvas:

- Existing spatial capture.
- Existing guidance.
- Recorded path where available.
- Small plan inset if it remains readable.

Persistent controls:

- Photo.
- Voice.
- Add moment.
- Finish.

Video can be accessible through the media action rather than becoming another competing primary button.

Status:

- Recording indicator.
- Elapsed time.
- Moment count.
- Tracking interruption when applicable.

7.4 Capture constraints

Validate whether the existing camera pipeline supports simultaneous scanning and photo/video capture.

If it does not:

- Use a deliberate capture interruption or review-time attachment flow.
- Preserve the trail draft.
- Explain the transition.
- Do not add a second camera session that silently breaks tracking.

Audio may continue alongside capture only after verifying session compatibility.

8. Moments and media

8.1 Quick moment

Tapping Add moment:

1. Creates a draft moment.
2. Associates it with the current valid spatial position.
3. Opens a compact sheet.
4. Allows a label and note.
5. Returns to capture after saving.

Default labels are allowed:

- Moment 1.
- Photo moment.
- Voice moment.

Naming is optional during capture.

8.2 Photo

- Take a photo or attach one during review.
- Confirm successful attachment.
- Offer Retake.
- Do not require a caption.
- Preserve original image orientation.

If attached after recording, allow the user to choose a moment. Do not invent its spatial position from attachment time.

8.3 Audio

- Explicit start and stop.
- Visible recording indicator and duration.
- Clear saved confirmation.
- Playback during review.
- Microphone permission requested when first needed.
- No automatic transcription.

A route-level introduction recording may be unpositioned; moment audio belongs to a specific moment.

8.4 Video

- Short clips.
- Clearly show recording duration.
- Set an explicit maximum before implementation; recommended initial cap: 30 seconds.
- Permit preview and deletion.
- Do not autoplay with sound.
- Save locally, subject to available storage.

If in-session video threatens capture stability, support video attachment during review first.

8.5 Positioning failure

When tracking is insufficient:

- Save the content as an unplaced moment.
- Mark it clearly.
- Allow manual placement during review.

Never silently place it at the origin or a stale coordinate.

9. Finish, save, and review

Finish sequence

```
Finish
→ Stop active media recording
→ Persist available session data
→ Finalise capture
→ Show processing status
→ Open review
```

A captured trail must not depend on completion of the metadata form.

Review screen

Show:

- Cover or spatial preview.
- Editable title.
- Place.
- Optional floor.
- Optional introduction.
- Ordered moments.

- 3D/Plan preview.

Actions:

- Revisit trail.
- Edit moments.
- Done.

Status wording must distinguish:

- Draft saved.
- Processing.
- Saved on this iPhone.
- Processing failed; available data retained.

Do not call a trail fully saved before required assets are persisted.

Editing

The user can:

- Rename.
- Change introduction.
- Change place association.
- Change floor.
- Choose cover.
- Edit or delete moments.
- Attach media.
- Reorder moment presentation.

Keep original capture order internally even if presentation order changes.

10. Revisit experience

Two modes:

Moments

A guided presentation of the recorded experience.

- Cover and introduction.
- Ordered moment cards.
- Highlighted spatial positions.
- Photo display.
- Explicit audio/video playback controls.

- Previous/Next.
- Progress such as “Moment 2 of 6.”

Explore

A freely inspectable view.

- 3D/Plan toggle.
- Pan, rotate, and zoom as supported.
- Fit scene.
- Recorded path.
- Selectable moments.
- Optional path visibility toggle.

Mode continuity

- Selecting a moment highlights it in both views.
- Switching modes preserves selection.
- Exiting and returning during the same app session restores context.
- Audio stops or continues according to one consistent, documented rule; default to stopping when leaving the trail.

Presentation mode

Provide “Show trail” or a full-screen viewing action:

- Hide editing controls.
- Emphasise media and authored content.
- Keep navigation obvious.
- Exit clearly.

This supports showing someone the experience on the same phone without implementing network sharing.

11. Collection: Journal and Map

11.1 Journal view

A warm editorial collection.

- Featured recent trail.
- Compact rows for the rest.
- Cover, title, place, date, and media indicators.

- Clear local status.
- Search.
- Sort newest/oldest.

Avoid loading full spatial assets into collection cards.

11.2 Map view

A geographic index of places and trails—not an indoor positioning engine.

Zoom level	Display
World	Clusters with trail counts
Region/city	Place markers
Place	Selected-place sheet and trail list
Indoor	Explicit transition into saved spatial viewer

Requirements:

- Geographic anchors locate places.
- Indoor geometry remains in local coordinates.
- Missing location does not prevent recording or appearing in Journal.
- Offer Add location for unlocated trails.
- Do not place records at fabricated coordinates.
- Online basemap availability must be distinguished from locally available trail content.

11.3 Shared filters

- Text search.
- Year.
- Custom date range.
- Place.

Both views share the same filter state.

Show:

- Active-filter chips.
- Clear filters.
- Result count.
- Fit filtered results on Map.
- Helpful no-results state.

Define date filtering using the trail's recording date, not its last edit date.

12. Multiple trails at one place

Place collection

Displays:

- Place name and optional locality.
- Geographic anchor if supplied.
- Trail count.
- Date span.
- Trail cards.
- Start another trail here.

Trail independence

A new recording has:

- Its own ID.
- Its own capture/session frame.
- Its own moments and assets.
- A shared place association.

Do not merge independent captures merely because they use the same place.

Inspired-by relationship

Optional metadata:

- Source trail ID.
- Optional source moment ID.
- User-written relationship note.

This supports:

> “Another perspective on the courtyard.”

It does not:

- Duplicate source media automatically.
- Align coordinate frames.
- Establish a physically verified branch point.

Exact spatial branching is deferred.

13. Identity and heritage

Every trail should make authorship visible through:

- Creator name.
- Recording date.
- Introduction.
- Attributed notes and voice.

Avoid reducing identity to a badge or score.

Heritage-oriented copy must remain invitational:

> “What would you like someone to remember?”

Do not:

- Claim every recording is historically authoritative.
- Invent stories.
- Imply private local content has entered a permanent public archive.
- Promise future-generation access without preservation infrastructure.

14. Data and persistence

Core records

Record	Required fields
Trail	ID, schema version, creator ID, title, timestamps, lifecycle state, place ID, capture references, path reference, moment IDs
Place	ID, name, optional locality, optional geographic anchor
Moment	ID, trail ID, label, note, original sequence, presentation order, optional spatial position, media references
Media	ID, type, local path, creation time, duration where relevant, thumbnail reference, processing state
Profile	Local ID, display name, optional introduction and portrait
Coordinate frame	ID, session/segment reference, units, axis convention
Path sample	Timestamp, local position, tracking state, session segment

Persistence rules

- Store drafts before capture begins.
- Persist moments and path data incrementally.
- Save manifests atomically.
- Keep original assets separate from derived previews.

- Use relative asset references.
- Version the format.
- Generate thumbnails independently.
- Delete owned assets when a trail is deleted.
- Do not delete shared assets while another record references them.
- Recover incomplete states on launch.

Privacy

- No automatic uploads.
- No mandatory analytics.
- No network dependency for recording and reopening downloaded/local assets.
- Platform file protection configured deliberately.
- Backup eligibility documented.
- No unsupported encryption or recovery promises.

15. Lifecycle states

```
Draft
  → Recording
  → Finalising
  → Processing
  → Saved

Recording → Interrupted
Finalising/Processing → Recoverable failure
```

Required behaviours:

- Relaunch detects unfinished sessions.
- Preserve usable partial content.
- Explain which assets are available.
- Allow deletion of failed records.
- Retry processing where source data permits.
- Do not label partial success as complete success.

16. Accessibility and interaction quality

- Dynamic Type on text-based screens.
- VoiceOver labels for controls and media.
- Selection conveyed by more than colour.
- Reduced-motion support.
- No unexpected audio playback.
- Clearly visible recording states.
- Accessible playback controls.
- Destructive actions require confirmation.
- Large touch targets.
- Spatial features are not marketed as accessibility-grade navigation.

Motion

Use motion only to explain:

- Which moment belongs to which position.
- Which trail is selected.
- Where a sheet originated.
- How a saved journey progresses.

No animation should delay saving or obscure capture guidance.

17. Architecture

```
Trails UI
  ├── Collection: Journal / Map
  ├── Capture
  ├── Review
  ├── Revisit
  └── Profile / Settings

Domain services
  ├── TrailRepository
  ├── PlaceRepository
  ├── CaptureCoordinator
  ├── MomentService
  ├── MediaService
  ├── ReplayController
  └── PreviewGenerator

Existing Extensa implementation
  ├── Spatial capture
  ├── Tracking
  ├── 3D renderer
  └── Plan renderer
```

```
Local persistence
  ├── Metadata index
  ├── Versioned manifests
  └── Assets
```

Future integrations attach through separate adapters:

- Export/import.
- Logos storage.
- Account sync.
- Public directory.
- AI processing.

Do not make present-day capture depend on those adapters.

18. Release priorities

P0 — Must work

- Capture.
- Path preservation.
- Text, photo, and audio moments.
- Save/reopen.
- Journal collection.
- Place grouping.
- Review/edit.
- 3D/Plan exploration.
- Moment sequence.
- Local deletion.
- Clear failure states.

P1 — Complete the v0.2.0 experience

- Geographic map.
- Date/year filters.
- Search.
- Presentation mode.
- Cover selection.
- Short-video attachment.
- Storage information.

P2 — Only after core acceptance

- Video during active scanning.
- Inspired-by relationships.
- Path-reveal animation.
- Advanced comparison of aligned trails.
- Additional styling and transitions.

If time runs short, release an honestly scoped build rather than leaving nonfunctional controls in the interface.

19. Acceptance scenarios

A. Personal memory

1. Record an indoor trail.
2. Add a photograph.
3. Leave a spoken observation.
4. Add a written moment.
5. Finish and save.
6. Close the application.
7. Reopen without internet.
8. Revisit the capture and all media.

Expected: content, order, and spatial associations remain intact.

B. Multiple perspectives

1. Record two trails.
2. Associate both with the same place.
3. Open the place collection.
4. Open each independently.

Expected: no overwritten recordings or falsely merged geometry.

C. Geographic collection

1. Add confirmed geographic anchors to several places.
2. Open Map.
3. Filter by year.
4. Select a place.
5. Open its trail.

6. Return to Map.

Expected: filter, selection, and map context remain consistent.

D. Show someone

1. Open a saved trail.
2. Enter presentation mode.
3. Play an audio moment.
4. Advance to a photo.
5. Switch to Plan.
6. Exit presentation.

Expected: the experience is understandable without editing controls or developer explanation.

E. Interrupted recording

1. Interrupt an active session.
2. Relaunch.
3. Inspect the recovered record.

Expected: available content is preserved and incompleteness is stated clearly.

F. Media compatibility

1. Record audio during scanning.
2. Attempt supported photo/video flows.
3. Continue capture.
4. Reopen the result.

Expected: no silent loss of tracking, media, or capture state.

20. Definition of done

Trails v0.2.0 is ready when a person can:

> Capture a place, leave meaningful moments, preserve the journey privately, find it again through their journal or map, and revisit—or show—the experience as a coherent personal story.

The 3D model is the setting. The path is the sequence. The moments are the meaning. The person is the author.